

AI-Agent for Trace analysis and Bottleneck Mitigation in Distributed Training Systems

Abstract

This report presents a complete study on designing an AI-Agent integrated with Ollama and Astra-sim insights to automatically analyze astra-sim outputs , identify bottlenecks and recommend optimization strategies.

The project probes three representative models mentioned below under multiple GPU scaling levels (1,2,4,8) and network topologies (Ring , Full Connected , Tree) :

- ResNet-50 (compute-bound)
- BERT-Base (balanced)
- DLRM (communication-bound)

The agent processes a .log file from your astra-sim output and produces a multipart analytical report that includes performance metrics , root cause identification, recommendation and forecasting the gains from those recommendations. To train and test the capabilities of the agent's processing and inference capabilities , controlled bottlenecks were created using perturbations in the system , network and communicator group configuration files in the Astra-sim input configuration files.

The experimental results show that the AI-agent's recommended changes led to a 16% reduction in the overall wall time , a 20% decrease in communication time and a 27% increase in GPU utilization along with improvement in throughput of approximately 24%.

1 . Introduction

The immense growth in the model complexity and the dataset size has made things like distributed learning the backbone of the modern AI infrastructure. However , the increase in the number of GPUs and nodes has caused the performance scaling of such models to often reach plateaus which are often far below ideal due to network congestion ,synchronization delays and load imbalance .

Traditional profilers such as Tensorboard or the Pytorch profiler provide detailed event traces but they rely heavily on manual interpretation . When the system scales , the sheer volume of

these logs becomes overwhelming which makes it tough to correlate different metrics like network-level , system-level , and application-level to detect performance detection patterns. Here Astra-sim comes into play , it is a co-simulation framework which addresses this by enabling synthetic experimentation across different configuration files in which we can change certain parameters such as topology , bandwidth , link-latency or bandwidth etc . Yet human analysis is needed to find meaning in those log files which output as a result of running such simulation on Astra-sim. This gap shows the need for the introduction of an AI-agent which is capable of parsing such simulation logs , identifying bottlenecks and generating suggestions to optimize the model.

The AI-agent developed in this project functions as a performance analyst, basically a tool that takes input the log files , classifies the bottleneck type using pattern and metric based reasoning then generates interpretable recommendations for the said model.

Objectives can be put

2.Methodology

The methodology consists of four main steps , each step contributes to a unique layer of data , learning and interpretation :

1. Trace simulation and data generation
2. Bottleneck variation while changing the configurations
3. AI - Agent log analysis framework
4. Result validation through re-simulation

2.1 Trace Collection and Preprocessing

For each model ResNet, Bert and DLRM a detailed trace was collected using Pytorch profiler and Kineto Traces. These collected traces included GPU kernel timestamps, communication events between GPUs and markers for synchronization . These were then converted to Chakra execution traces with the help of Chakra forming one of the inputs for the Astra-sim simulations.

To ensure diversity following parameters were varied ;

1. GPU Count: 1, 2, 4, 8
2. Batch Size: 8, 16, 32, 64
3. Gradient Accumulation Steps: 1, 2, 4
4. Network Topology: Ring, Tree, Fully Connected

The core innovation of this project lies between the Reasoning Layer of the AI-agent , which bridges the gap between qualitative data collected from Astra-sim with qualitative and good insights about the bottlenecks in the system .

Each Astra-sim execution generates .log files containing the per-rank metrics such as wall time and communication time . The AI-agent starts by parsing this file and generating and extracting key metrics such as below (_Given that per GPU batch size is also given as input) :

Metric	Derivation Formula
1. Average Wall Time ($\bar{T}_{Wall}$)	$\frac{1}{N} \sum_{i=1}^N T_i$
2. Average Exposed Comm Time ($\bar{T}_{Comm}$)	$\frac{1}{N} \sum_{i=1}^N C_i$
3. Inferred Avg. Compute Time ($\bar{T}_{Compute}$)	$\bar{T}_{Wall} - \bar{T}_{Comm}$
4. Communication Fraction (CF)	$\frac{\bar{T}_{Comm}}{\bar{T}_{Wall}}$
5. Compute Fraction / GPU Utilization (CompF)	$\frac{\bar{T}_{Compute}}{\bar{T}_{Wall}}$
6. C2C Ratio (C2C)	$\frac{\bar{T}_{Comm}}{\bar{T}_{Compute}}$
7. Rank Imbalance (RI)	$\frac{Max(T_i) - Min(T_i)}{\bar{T}_{Wall}} \times 100\%$
8. Throughput (Thr)	$\frac{GBS}{\bar{T}_{Wall}/S}$

The reasoning and Classification Layer of the AI-agent is implemented using a rule-guided, context aware reasoning system built on the Llama3.2 Ollama model . Instead of purely statistical analysis it imploys pattern-based classification which is then developed by contextual prompts

Condition ID	Metric Conditions	Inference	Interpretation / Rationale
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R1	<code>comm_fraction > 0.45</code> and <code>imbalance > 5%</code>	Network Congestion	High comm relates to uneven rank timing which further indicates congestion or link saturation.
R2	<code>compute_fraction > 0.70</code> and <code>gpu_util < 60%</code>	Compute Inefficiency	GPU is not fully occupied meaning kernel scheduling or a memory latency bottleneck.
R3	<code>comm_fraction > 0.35</code> and <code>compute_fraction > 0.55</code>	Mixed Bottleneck	
R4	<code>overlap_ratio < 0.25</code> and <code>comm_fraction > 0.40</code>	Poor Overlap Efficiency	Communication is not overlapped with computation; it is likely a synchronization bottleneck.
R5	<code>rank_imbalance > 7%</code>	Straggler-Induced Synchronization Delay	If one or more ranks finish significantly later it is often due to uneven workload distribution.
R6	<code>throughput_stdev / throughput_mean > 0.10</code> and <code>gpu_util_var > 10</code>	Temporal Instability	If Throughput fluctuates between iterations it suggests transient congestion or jitter.
R7	<code>compute_fraction > 0.75</code> and <code>c2c_ratio < 0.2</code>	Arithmetic Bound (FLOP-Limited)	Meaning GPU is fully busy and the performance gated by arithmetic throughput.
R8	<code>comm_fraction > 0.55</code> and <code>overlap_ratio > 0.5</code> but <code>gpu_util < 65%</code>	Pipeline Imbalance	Communication overlap exists but compute remains underutilized which indicates unbalanced pipeline stages.
R9	<code>scaling_efficiency < 70</code> and <code>comm_fraction > 0.5</code>	Bandwidth Saturation	Adding GPUs increases communication cost faster than the savings on computation.

These rules are evaluated hierarchically and in the case of multiple triggers the agent then uses the below rule , ensuring that the final classification of the bottleneck reflects the

dominant rule cause rather than triggering every contributing issue:

Network Congestion > Mixed > Compute > Synchronization > Instability

This ensures that the root causes are addressed before the secondary effects which prevents redundant or counterproductive tuning of the models.

The AI-agent’s classification was derived empirically from the extensive analysis of Astra-sim outputs guided by prior research and reading of certain research papers . Over 60 simulations across the three different models (Resnet50 , DLRM , Bert-based) , GPU count and different network topologies each giving different performance on communication fraction , compute fraction , rank imbalance & GPU utilization . By correlating the metric of these models and their bottlenecks threshold ranges were identified that separated one bottleneck from the other . In essence the above rules were not chosen randomly; these were nothing but data-driven thresholds grounded with measurable system behaviour ensuring that the reasoning is technically defensible .

3. Recommendation mapping and Optimization Strategies :

Once the AI-agent classifies the dominant bottleneck type it switches to a custom recommendation mapping that aligns the detected bottleneck type with a recommendation map that aligns with detected system problems with a set of corresponding recommendation or actions . The below map was designed using a combination of domain informed heuristics and empirical evidence which was collected from multiple simulation interactions . Each optimization action targets a specific layer of the distributed training stack - network topology , communication protocol , batch scheduling or parallel strategies depending on where the agent locates the inefficiencies.

Bottleneck Type	Recommended Interventions	Expected Gain (%)	System Layer Affected
Network Congestion	Topology switch, accumulation, compression	15–25	Network
Poor Overlap	Async ops, prefetch, graph reorder	8–15	Execution pipeline

Compute Inefficiency	Batch size ↑, kernel fusion, FP16	10–20	GPU/Kernel
Straggler Delay	Data rebalancing, adaptive load balancing	5–10	Scheduling
Bandwidth Saturation	Hybrid parallelism, comm compression	15–20	Communication

Sensitivity Analysis : Batch Size and Gradient Accumulation

To ensure that the agent gives appropriate optimizations for batch size and gradient accumulation . Two key experiments were designed to see , actually how performance changes or responds to training configuration changes :

A. Batch Sweep

The compute to communication ratio is directly affected by batch size , improving arithmetic intensity but may saturate the GPU memory . The agent learned that the throughput linearly up to a batch-size of 64 beyond which the memory limits reduced gains .

Batch Size	Compute Fraction (%)	Comm Fraction (%)	Throughput (samples/s)
8	53	47	540
16	58	42	670
32	62	38	760
64	65	35	830

B. Gradient Accumulation Steps

The gradient accumulation reduces the communication frequency by merging multiple backward passes before the synchronization. It was further observed that increasing accumulation above 4 showed diminishing returns due to the delayed parameter updates.

Accum Steps	Comm Reduction (%)	Wall Time (ms)	Throughput	Recommendation
1	—	265	665	—
2	18	240	770	Moderate gain
4	25	228	825	Optimal setting

The AI-agent processed astra-sim logs from Resnet-50 , Bert-base and DLRM across different numbers of GPUs extracting compute , communication and overlap metrics. The inferences were governed by a rule based reasoning mapping which is from R1 -> R10 , which was designed to link the bottleneck types to metric patterns . Each condition was helpful in evaluating metric thresholds such as communication fraction , rank imbalance or overlap ratio & which further guided the classification . explanation and strategy for the AI-agent .

4. Results

This section summarizes everything that our system was able to extract, compute, and understand from the Astra-Sim traces. The results combine three layers of analysis:

1. **The raw per-rank timings** directly parsed from the `.log` files
2. **The computed system-level performance metrics** (communication fraction, compute fraction, GPU utilization, imbalance, scaling efficiency, etc.)
3. **The higher-level interpretation** done by the Rule Engine (R1–R10) and the AI-Agent

Together, these pieces show not just *what* is happening inside the simulation, but also *why* it is happening and *how* the system behaves as the number of GPUs, topology, and workload change. To make the evaluation realistic, we simulated three representative deep-learning workloads—**ResNet-50**, **BERT-Base**, and **DLRM**. Each model was run on **1, 2, 4, and 8 GPUs**, allowing us to capture the scaling behavior, communication overhead patterns, and sensitivity to network topology. The analyzer then automatically computed:

- per-rank wall time / communication time
- compute and communication fractions
- C2C ratio and overlap ratio
- rank imbalance percentage
- throughput (based on global batch size and iteration time)
- GPU utilization and scaling efficiency
- temporal instability signals

Once these metrics were computed, the Rule Engine checked each run against the ten diagnostic rules (R1–R10). This produced a structured summary of which bottlenecks were present—for example, network congestion, poor overlap, compute inefficiency, pipeline imbalance, and so on.

Finally, the **AI-Agent** took all this structured data and generated deeper explanations:

- What the dominant bottleneck really means
- Why it happened in this configuration
- What architectural or algorithmic factors caused it
- What to fix first, and why
- What risks appear if we ignore the issue

The rest of this Results section presents:

- A clear breakdown of all parsed metrics per rank
- The computed performance metrics for each workload and GPU count
- Which rules were triggered and what bottlenecks they indicate
- The AI-Agent’s interpretation and recommendations
- Visualization-based insights (heatmaps, rank imbalance, time pie charts)
- Comparisons between models and GPU scales
- Sensitivity results for batch size and gradient accumulation

Overall, these results demonstrate that the analyzer is able to capture subtle performance issues, classify them using deterministic rules, and explain them using an LLM in a human-readable and actionable format.

Rule ID	Model	Result (Bottleneck Type → Mitigation → Outcome)
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R1	DLRM (8 GPU, Ring)	Bottleneck: Network Congestion (comm fraction 0.54, imbalance 7.5%) Mitigation: Switch to Fully Connected topology + increase gradient accumulation to 4 Result: Comm time ↓ 22%, throughput ↑ 25%, scaling efficiency ↑ to 89%.
R2	ResNet-50 (1 GPU)	Bottleneck: Compute Inefficiency (kernel under-utilization) Mitigation: Increase batch size from 32 to 64 Result: GPU utilization ↑ from 58% to 91%, throughput ↑ 13%, compute fraction balanced.
R3	BERT-Base (8 GPU, Tree)	Bottleneck: Mixed (Comm + Compute) Mitigation: Use Tree topology + set gradient accumulation = 2 Result: Throughput ↑ 17%, balanced compute-comm overlap.
R4	BERT-Base (4 GPU, Ring)	Bottleneck: Poor Overlap Efficiency (overlap ratio 0.22, comm fraction 0.40+) Mitigation: Enable async overlap + switch to Tree topology Result: Overlap ↑ to 47%, iteration latency ↓ 12%, throughput ↑ 14%.
R5	DLRM (8 GPU, pre-opt)	Bottleneck: Straggler-Induced Synchronization Delay (imbalance 7.5%) Mitigation: Redistribute load + balance ranks Result: Imbalance ↓ to 2.3%, straggler ranks eliminated, sync delays removed.
R6	BERT-Base (4–8 GPU)	Bottleneck: Temporal Instability (throughput stdev/mean > 0.10, GPU util var > 10) Mitigation: Enable comm-compute overlap + increase gradient accumulation Result: Throughput variance ↓ from 11% → 5%, stabilized performance.
R7	ResNet-50 (4–8 GPU)	Bottleneck: Arithmetic Bound (FLOP-Limited; compute fraction > 0.75, C2C < 0.2) Mitigation: None required (baseline optimal) Result: Scaling efficiency is 98%, confirmed the compute-bound saturation and has excellent GPU utilization.

R8	DLRM (4 GPU, Ring)	Bottleneck: Pipeline Imbalance (comm > 0.55, overlap > 0.5, GPU util < 65%) Mitigation: Reduce per-GPU batch size + tune overlap synchronization Result: GPU util ↑ from 60% → 71%, throughput ↑ 12%, balanced pipelines.
R9	DLRM (4 GPU, Ring)	Bottleneck: Bandwidth Saturation (scaling eff < 70, comm > 0.5) → Mitigation: Adopt Fully Connected topology Result: Scaling eff ↑ to 89%, throughput ↑ 25%, network links unsaturated.

The Key Insights from the Results Are:

- **Communication Topology Matters:** The structure of node connections, whether Ring or Tree, directly influences congestion and throughput.
- **Compute–Communication Balance:** Both phases must be optimized together; otherwise, one limits the other.
- **Scaling Doesn’t Always Equal Efficiency:** With more nodes, imbalance and overhead can be introduced unless tuned properly.
- **AI-Assisted Diagnosis Is Valuable:** AI simplified root cause identification through a more interpretable set of recommendations, compared to manual analysis.
- **Batch Size Sensitivity:** Optimal batch size depends on node count and topology; improper sizing can worsen communication overhead and GPU utilization.
- **Gradient Accumulation is Effective:** Accumulating gradients locally reduces communication frequency and improves throughput.
- **Hierarchical topologies scale better:** Tree or fully connected networks are always superior to ring-based designs for multi-node arrangements.

5. Discussion

The experimental results validate the robustness and interpretability of the proposed AI-Agent for the trace analysis of Astra-Sim. The agent has rightly identified dominant bottlenecks; suggested corrective actions through a systematic evaluation across ResNet-50, BERT-Base, and DLRM on 1, 2, 4, and 8 GPUs; and validated the improvements through re-simulation. The discussion below outlines the major observations, the nature of the bottlenecks, and the broader insights gained.

5.1. Agent Reasoning and Rule-Based Precision

The performance of the AI-Agent was not based on obscure machine learning heuristics but on a transparent, metric-driven reasoning engine.

Every inference directly mapped to one or more predefined rules (R1–R10) that correlated a specific metric threshold, such as communication fraction, rank imbalance, overlap ratio, GPU utilization, or scaling efficiency, with a bottleneck class: Compute, Communication, or Mixed.

This approach allowed the agent to perform pattern-based classification supported by explainable traces of decisions.

For example:

- R1 (Network Congestion) and R9 (Bandwidth Saturation) provided reliable detection for communication-heavy issues in DLRM.
- R2 (Compute Inefficiency) and R7 (Arithmetic Bound) characterized compute-limited behavior in ResNet.
- R3, R4, R6, and R10 highlighted synchronization and overlap inefficiencies in BERT.

By enforcing these logical mappings, the agent's results became interpretable, reproducible, and verifiable — aligning closely with expert-level performance analysis.

5.3 Iteration-Level and Scaling Observations

The agent's fine-grained analysis provided iteration breakdowns, per-rank utilization, and sensitivity sweeps for batch size and gradient accumulation:

- Increasing batch size improved the compute utilization for compute-bound models (ResNet).
- Increasing the accumulated gradient reduced the communication overhead for bandwidth-limited cases (DLRM, BERT).
- The AI-Agent measured predictable scaling patterns across GPU scales—near-linear for compute-bound models and sublinear for communication-bound ones (1 → 8).
- Per-rank utilization variance decreased significantly after optimization to indicate that network congestion and synchronization delays had been greatly relieved.

The combination of these analyses confirmed that the AI-Agent could not only detect bottlenecks but also quantify the effect of corrective actions, turning simulation metrics into actionable intelligence.

5.6. Key Takeaways

- Each of the identified bottlenecks had a quantitative and measurable improvement post-mitigation.
- The rule-based inference framework furnished explainable AI for performance analysis by closing the gap between simulation metrics and system design decisions.
- Further, this methodology can be integrated into future performance monitoring pipelines to automate diagnosis for large-scale training environments.

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